

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

27th May 2014

Subject CT1 – Financial Mathematics

Time allowed: Three Hours (10.30 – 13.30 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** An investor pays ₹ 2,000/- every half-year in advance into a 15-year savings plan. Calculate the accumulated fund at the end of the term, if the interest rate is 8% per annum convertible monthly for the first 7 years and 6% per annum convertible half-yearly for the remaining 8 years. [4]
- Q. 2)** A continuous cash flow measured in years is to be paid at the rate $p(t) = 4 + 0.5t$, $0 \leq t \leq 8$. The force of interest applicable during the period is: $\delta(t) = 0.04 + 0.005t$, $0 \leq t \leq 8$. Find the accumulated value of the cash flow at the end of the 8th year. [5]
- Q. 3)** i) Explain the term “Equation of value” (1)
- ii) An investor deposits ₹ 4000/- in a bank. He withdraws this amount in ten level annual payments starting a year after the deposit. Along with the tenth annual payment, the investor will also be refunded a certain percentage of the deposit amount. Calculate the percentage, given that the annual rate of interest is 7% per annum. (3)
- [4]
- Q. 4)** i) A 90-day government bill was bought by an investor for a price of 95 per 100 nominal. After 60 days the investor sold the bill to a second investor for a price of 98.42 per 100 nominal. The second investor held the bill to maturity when it was redeemed at par. Compare and comment on the annual effective rate of interests earned by the first and the second investor. (3)
- ii) A 91-day government bill is discounted at a simple rate of discount of 9% per annum. Calculate the annual effective rate of interest earned on the investment. (3)
- [6]
- Q. 5)** The one-year annual effective interest rate is currently at 10%. It is expected to be at 9% after a year, 8% after two years and 7% after three years.
- i) Based on the expectations theory, calculate the gross redemption yields (spot rates of interest) from 1 year, 2 year, 3 year and 4 year zero coupon bonds. (4)
- ii) The price of a coupon paying bond is calculated by discounting each of the payments from the bond at the zero-coupon bond yields in (i). Calculate the gross redemption yield of a bond that is redeemed at par in exactly four years and pays a coupon of ₹7 per annum annually in arrears. (3)
- [7]
- Q. 6)** i) Define the running yield on investment in ‘property’. (1)
- ii) Explain, with reasons, whether the running yield on investment in ‘property’ is normally higher or lower from that on ordinary shares. (2)
- [3]

- Q. 7)** A loan of nominal amount ₹ 5,00,000 is to be issued by a borrower which offers coupons payable monthly in arrears at a rate of 9% per annum. After the date of issuing the loan, the capital is to be redeemed at 109 for 100 nominal on any of the coupon dates between 20 and 25 years (both inclusive). Redemption date is to be decided by the borrower.

An investor with an applicable income tax rate of 10% per annum and capital gains tax rate of 20% per annum is considering the purchase of the total loan on the date of the issue. Calculate the price which the investor should pay to ensure a net effective yield of not less than 8% per annum.

[8]

- Q. 8)** i) Define the terms - Futures Contract and Forward Contract (2)

- ii) An investor had entered into a long forward contract for a security eight years ago. The contract is due to mature in twelve years from now. The price of the security under the contract was ₹ 75 eight years ago and the price today is ₹ 175. The risk-free rate of interest can be assumed to be 8% per annum throughout the contract period of 20 years.

With 'no arbitrage' assumption, arrive at the value of the contract today, if the dividends of ₹ 15 and ₹ 20 are payable on the security in four years and eight years respectively from now.

(3)

- iii) A one-year forward contract was issued on 1st January 2014 on a security with a price of ₹ 1200 at that date. On 1st April 2014 and on 1st October 2014, payments of ₹ 100 per security are expected. On 1st January 2014, the 3-month and the 9-month spot rates are 8% and 9% per annum effective respectively. The risk-free rate for the period of the contract is 9% per annum effective.

Calculate the forward price at issue, stating any assumptions.

(4)

[9]

- Q. 9)** A student actuary has invested an amount of ₹ 1,25,000 in a corporate deposit which pays interest at the end of each year. The practice adopted by the student actuary is to always reinvest interest earned on the investment in the same corporate deposit. The rate of interest is declared at the beginning of each year which remains unchanged during the year. The rate of interest declared in any one year is independent of the rate declared in any other year.

At the beginning of the first year, the annual effective rate of interest declared is expected to be one of 4%, 5% or 6% with equal probability.

At the beginning of the second year, the annual effective rate of interest declared is expected to be either 8% with 40% probability or 5% with 60% probability.

At the beginning of the third year, the annual effective rate of interest declared is expected to be either 7% with 90% probability or 5% with 10% probability.

- i) Derive the expected accumulated amount of the investment at the end of three years. (5)
 - ii) Calculate the probability that the accumulated amount of the investment is more than ₹ 1,53,000 at the end of three years. (3)
- [8]**

Q. 10) An investor is analysing two investment opportunities. The yearly cash flows associated with each of the investment opportunities are as given below:

Project X: An investment of ₹ 39,90,000/- is made to purchase a residential house at time zero and an additional investment of ₹ 20,00,000/- is made towards refurbishment at the end of the second year. In return, the investor is expecting to get a rental income of ₹ $11,20,000 \times 1.02752^t, t = 0 \text{ to } 14$, per annum from the end of 5th year for 15 years. The rental income is received in arrears.

Project Y: Under this project, the investor pays a regular payment of ₹ 2,00,000/- per annum for 12 years made at the start of each year. In return, the investor receives an increasing amount of A, 2A, 3A and so on annually from the end of the 12th year for the next 15 years where A= ₹ 1,20,000/-.

- i) Find the net present value (NPV) of the project X at an annual effective interest rate of 12% per annum. (6)
 - ii) Find the internal rate of return on the project X without making any further calculations and ignoring any NPV less than 1000. (1)
 - iii) Find the NPV for the Project Y at an annual effective interest rate of 11% per annum. (4)
 - iv) Find the value that A would need to be for the internal rate of return of the project Y to be the same as that of the project X. (3)
- [14]**

Q. 11) An investment trust has two liabilities:

1 - Annuity payments of ₹ 1,00,000 per annum is to be paid annually in arrears for next 12 years; and

2 - A lump sum payment of ₹ 3,00,000 is to be paid in twenty years.

The investment trust wishes to invest in two fixed-interest securities in order to immunize its liabilities. Security X has a term to redemption of 7 years with a coupon rate of 7% per annum. Security Y has a term to redemption of 22 years with a coupon rate of 5% per annum. Both the securities pay coupons annually in arrears and are redeemable at par.

- i) Calculate the present value of the liabilities at a rate of interest of 6% per annum effective. (2)
 - ii) Calculate the discounted mean term of the liabilities at a rate of interest of 6% per annum effective. (4)
 - iii) Calculate the nominal amount of Security X and Security Y to be purchased so that both the present value and discounted mean terms of assets and liabilities are equal. (7)
 - iv) Without doing any further calculations, if the conditions in (iii) are fulfilled, comment on whether the investment trust is likely to be immunized against small, uniform changes in the rate of interest. (2)
- [15]**

Q. 12) A person takes out a loan for over 15 years and makes monthly repayments in arrears. The first payment is ₹ 100 and the remaining payments increase by ₹ 5 each time until the payments reach ₹ 700. The remaining payments are ₹ 700/- each. The annual effective interest rate is 8% per annum.

- i) Calculate the amount of the loan. (6)
- ii) Calculate the amount of the capital outstanding after the payment of the 24th instalment. (4)
- iii) Comment on the amount of the capital due in the 25th instalment. (3)
- iv) At the end of the 2nd year (after the payment of the 24th instalment), the person decided to reschedule the loan. As per the rescheduled terms the balance repayment is made in arrears over 10 years and the rate of interest is 6% per annum. The repayment is made by $X(1.01923)^t$ where $t=0$ to 9. Find the value of X. (4)

[17]
